3.1 Technical principles

3.1.1 Materials and their applications

<table>
<tr><th>Content</th><th>Potential links to maths and science</th></tr>
<tr><td>Students are expected to be able to name specific materials for a wide range of applications.

They must also be able to provide detailed and justified explanations of why specific materials and combinations of materials are suitable for given applications, with reference to:
• physical and mechanical properties (working characteristics)
• product function
• aesthetics
• cost
• manufacture and disposal.</td><td>Understand the appropriate use of materials including polymers, composites, woods and metals based on their physical and working characteristics such as:
• malleability
• toughness
• hardness
• resistance to corrosion and degradation
• thermal conductivity
• electrical conductivity.

Calculation of quantities of materials sizes and costs.</td></tr>
</table>

Classification of materials

<table>
<tr><th>Content</th><th>Potential links to maths and science</th></tr>
<tr><td>Students should know and understand the classifications of the following materials and be able to name examples that belong to each category:
• metals (ferrous, non-ferrous, alloys)
• woods (hardwoods, softwoods, manufactured boards)
• polymers (thermoplastics, thermoset polymers, elastomers)
• papers and boards
• composites
• smart materials
• modern materials.</td><td></td></tr>
</table>

Methods for investigating and testing materials

Content	Potential links to maths and science
Students should be able to describe how workshop and industrial tests are set up and what will be tested, measured and compared, including: • tensile strength • toughness • hardness • malleability • corrosion • conductivity.	Analysis of data obtained from testing.